Frames and matrix designs

Uniform conditioning of large Paley-frame column subsets

Difficulty: extreme **Importance:** broadly interesting

Status: Open

Rating rationale: Uniform conditioning for this explicit arithmetic frame beyond square-root sparsity is a major deterministic sensing barrier linking number theory and numerical reconstruction.

For a prime $p \equiv 1 \pmod 4$, let Q be the nonzero quadratic residues modulo p, and set $N = (p+1)/2$. Define $\Phi_p \in \mathbb{C}^{N\times(p+1)}$ with row labels $\{0\} \cup Q$ and column labels $\{0, \dots, p-1, \infty\}$ by

$$(\Phi_p)_{0j} = p^{-1/2}, \qquad (\Phi_p)_{rj} = \sqrt{2/p}\exp(-2\pi i rj/p) \quad (r \in Q,\ 0 \le j < p),$$

and $(\Phi_p)_{0,\infty} = 1$, $(\Phi_p)_{r,\infty} = 0$ for $r \in Q$. All columns have Euclidean norm one.

Do constants $0 < \varepsilon < 1/2$, $c > 0$, $C < \infty$, and p_0 exist such that, for every prime $p \ge p_0$ with $p \equiv 1 \pmod 4$ and every column subset S with

$$1 \le |S| \le \lfloor cp^{1/2+\varepsilon} \rfloor,$$

the column submatrix satisfies

$$\kappa_2((\Phi_p)_S) = \frac{\sigma_{\max}((\Phi_p)_S)}{\sigma_{\min}((\Phi_p)_S)} \le C?$$

The condition number is infinite when the columns are dependent. Constants must be independent of both p and S.

This is the bounded-condition-number formulation of the Paley-frame conjecture in Randomstrasse 101, Conjecture 29. It asks for uniform conditioning beyond the square-root sparsity scale, rather than conditioning of a random subset. The explicit frame would provide deterministic measurement matrices with stable sparse least-squares subproblems. The general deterministic restricted-isometry construction problem does not require this particular matrix, so neither problem duplicates the other.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse 101: Open Problems of 2025*, arXiv:2603.29571 (2026), Conjecture 29 and the preceding Paley construction. Paper.
2. A. S. Bandeira, M. Fickus, D. G. Mixon, and P. Wong, *The road to deterministic matrices with the restricted isometry property*, Journal of Fourier Analysis and Applications 19 (2013), pp. 1123–1149. The Paley ETF discussion supplies the original restricted-isometry motivation. Paper.
3. S. Satake, *On the restricted isometry property of the Paley matrix*, Linear Algebra and its Applications 631 (2021), pp. 35–47. Its main implication is conditional on a Paley-graph conjecture. Paper.

Status check — 2026-09-10

Searched "Paley matrix RIP conjecture 2025 2026", "Paley ETF condition number square root bottleneck", and the exact Satake title; checked the 2026 restatement and current arXiv records. No unconditional proof or counterexample was found. Conditional graph-discrepancy implications do not settle the quantifiers above.

Audit update (2026-09-10): Rechecked the 2026 Conjecture 29 and searched for later Paley ETF results. The exact uniform condition-number target remains explicitly conjectured; no unconditional resolution was located. This is a bounded literature check, not a proof that no solution exists.